Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities Minneapolis, MN

Bachelor of Science in ***Mathematics*** • ***Computer Science Minor***

Expected May 2028

- **GPA:** 4.0/4.0 **| Honors:** Dean's List, Fall 2025, Spring 2026 · Tau Sigma Honor Society
- **Relevant Coursework:** Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists and Engineers I & II (A), Introduction to Programming Concepts (Python, A), Introduction to Statistical Analysis (A)

RESEARCH & INDUSTRY EXPERIENCE

Low-Carbon Transition & Industrial Monitoring Platform **Beijing / Shanghai, China**

Faculty-led Researcher (research phase) → Algorithm Engineer (industry phase), Shanghai Huaruizhongxin Technology Co., Ltd.

Oct 2024 - May 2025; Aug 2026 - Sep 2026

- Contributed to a faculty-led low-carbon transition project for China's steel industry, supporting a proposal and implementation plan for national ministry consideration through data collection, database construction, and quantitative analysis.
- During the industry implementation phase, contributed to an energy and carbon-emissions monitoring platform by designing a Python prototype for industrial time-series data quality governance, translating meter/point metadata and quality flags into rules for cumulative, instantaneous, consumption, enumeration, and statistical measurements.
- Implemented historical and real-time cleaning for range, constant, trend, missing-data, and communication anomalies; used same-time historical profiles for outage imputation and cumulative-to-consumption attribution; validated multi-day data with plots and regression tests, then packaged the work with Git version control.

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota Minneapolis, MN

Undergraduate Research Assistant & UROP Fellow, Advisor: Prof. Yan Liu Oct 2025 - Present

- Built **cdms_spotlight**, a multilingual natural-language query system for the SuperCDMS DQM database hosted at SLAC: server-side filtering scaling from ~1,300 to ~100,000 series, cuts on any of 10,929 ODB parameters per series, self-learning local NLP (TF-IDF + logistic regression); runs 24/7 as a Slack bot in the official collaboration workspace, pip-installable CLI; presented at the SuperCDMS Software Meeting. Jul 2026
- Built the **phonon pulse template pipeline** for 13 SNOLAB R4 detectors (12 phonon channels each): event selection around the 10.37 keV activation line (~120 GB raw traces), free-pretrigger 2-exponential fits with quality cuts; the 1×1 and PCA N×M ROOT templates serve as the R4 pulse templates for optimal-filter reprocessing. May–Jul 2026
- Built **BlueFors-Alert**: minute-level PostgreSQL sync of the 64 GB CS2 database to a Raspberry Pi; Slack monitor with dual IDLE/COLD thresholds and time-range plots of any sensor; extended to a liquid-nitrogen scale. Jul 2026
- Extended RuOx (R31279 & R31839) **calibration curves** to 5 mK–300 K (PCHIP + Mott VRH); the generated .340 files are now in use on the lab's BlueFors dilution refrigerator. May 2026
- Built a **Python pipeline** for LED calibration data from 11 cryogenic Ge channels (5,200+ events, 10+ GB raw): baseline subtraction, SNR-based glitch detection, peak alignment, exponential fitting. Oct 2025

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow Beijing, China

Team Leader Oct 2024 - Jun 2025

- Led a student team to design **a billiards rest** that combines the high and low bridge heads into one;
- Took it from CAD model to 3D-printed prototypes, iterating through player testing and market research; received **national project approval with seed funding**; **Chinese utility patent** under CNIPA review.
- Invited to present at the **Tsinghua X-Lab & Kering Entrepreneurship Roadshow** to industry leaders.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **1st Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **1st Prize**, National College E-Commerce "Innovation, Creativity & Entrepreneurship" Challenge, Campus Level | Team Leader
- **3rd Prize**, China International College Students' Innovation Competition, Campus Level (2025) | Team Leader
- **2nd Prize**, NECCS National English Competition for College Students

ACTIVITIES & SERVICE

- **Special Olympics USA Games Minnesota 2026**, Tennis Division - Volunteer Ball Retriever | June 25, 2026
- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, real-time PostgreSQL sync, local NLP, Slack bot development, waveform template construction
- **Tools:** Python (NumPy, Pandas, SciPy, Matplotlib), scikit-learn, PostgreSQL, Java, ROOT, Git, Linux, Slurm, MSI HPC (Agate)
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)